Advancing machine learning for protein structure prediction in drug discovery

Specific Aims

Aim 1: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy [1].

Aim 2: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy [2].

Aim 3: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy [3].

Background

This proposal addresses machine learning for protein structure prediction in drug discovery, a problem with substantial significance [4]. Our preliminary data suggest a tractable path toward measurable improvement [5]. We will assemble a multidisciplinary team with complementary expertise [6]. The approach integrates established materials methods with a novel analytical pipeline [7].

Approach

The approach integrates established materials methods with a novel analytical pipeline [8]. Prior work has established a partial understanding, yet key gaps remain. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement.

Innovation

Rigorous, pre-registered analyses will guard against bias and support reproducibility.

Investigator Context

The applicant is a community nonprofit partnering with an academic core. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Budget: personnel $1,647,060; equipment $292,480; travel $69,708; indirect costs $317,744. Total requested support is $2,326,992 over 3 years at a R1 institution.

As one source notes, “This proposal addresses the target problem, a problem with substantial significance.” [1]

References

1. [1] https://doi.org/10.1145/3292500.3330701
2. [2] https://doi.org/10.1126/science.1259855
3. [3] https://doi.org/10.1145/3292500.3330701
4. [4] https://doi.org/10.1021/jacs.9b02765
5. [5] https://doi.org/10.1289/ehp.1104477
6. [6] https://doi.org/10.1289/ehp.1104477
7. [7] https://doi.org/10.1038/nature14539
8. [8] https://doi.org/10.1289/ehp.1104477